SECTION 1: DUAL-TRANSPORT ACTIVE/ACTIVE SD-WAN ARCHITECTURE

Section 1.1 - Dual-Transport Active/Active SD-WAN with Sub-Second Failover

Every healthcare clinic is engineered with a Primary 10Gbps Dedicated Optical Wavelength plus a Secondary Diverse Carrier-Ethernet/Fiber transport terminated on dual-homed SD-WAN edge appliances. Per-packet Forward Error Correction (FEC) and sub-second synthetic health probes guarantee zero session drops for Citrix/Epic Hyperspace transactions.

SECTION 2: 10G OPTICAL WAVELENGTHS & NEAR-NET LATERAL CONSTRUCTION

Section 2.1 - Dedicated 10Gbps MEF 3.0 Optical Transport & ITU-T G.652.D Fiber

All 45 clinics utilize non-oversubscribed 10Gbps MEF 3.0 Carrier Ethernet / OTN wavelengths terminated on Ciena 5170 or Nokia 7210 SAS-S NIDs with continuous hardware ITU-T Y.1731 / TWAMP-Light optical performance telemetry.

Section 2.2 - On-Net vs. Near-Net Fast-Track Lateral Construction (<150m)

Across the 45 Apex Health clinics, 43 locations (95.6%) are lit On-Net buildings with existing single-mode fiber risers and reserved 10Gbps SFP+ ports. Only two locations require short Near-Net lateral fiber extensions (<150m): Site #28 (Wenatchee Valley Specialty Clinic, 82m directional micro-trenching from splice closure SC-QCY-028, CAPEX $1,820) and Site #33 (Hood River Gorge Medical Center, 118m conduit pull from splice closure SC-PDX-033, CAPEX $2,380). Total civil CAPEX of $4,200 is completed within 11 business days and amortized over 36 months.

Section 2.3 - SR-MPLS TI-LFA Core Optical Protection (<50ms)

Our regional DWDM backbone connecting Seattle, Bellevue, Tacoma, Quincy, Spokane, Portland, Eugene, Boise, and Sacramento runs Segment Routing (SR-MPLS) with Topology-Independent Loop-Free Alternates (TI-LFA) and optical roadm protection switching under 50ms.

Section 2.4 - Dual-Entry & Diverse Last-Mile Engineering

Regional medical centers utilize physically diverse north/south or east/west underground conduit building entries separated by >=25 meters. For Site #28 (Wenatchee) and Site #33 (Hood River), primary underground fiber entries are complemented by secondary alley vault entries and active 5G/LEO backup.

SECTION 3: HEALTHCARE QoS, BFD TIMERS & CUSTOM BGP ROUTING

Section 3.1 - Healthcare Hierarchical QoS (H-QoS) Traffic Engineering

Hardware H-QoS allocates 40% strict low-latency priority queuing (DSCP EF/CS5) for Epic EHR and Tele-ICU real-time streams, 35% assured forwarding (AF41) for PACS/DICOM medical imaging, and 25% for enterprise clinical applications.

Section 3.3 - High-Availability Dual-CPE Branch Design

All hospitals, tertiary surgical centers, and Near-Net regional hubs include dual redundant SD-WAN appliances in active/active stateful HA with dual power supplies and 8-hour battery backup.

Section 3.4 - Hardware-Assisted BFD (<45ms) & Custom BGP Community Tagging

Nokia 7750 SR PoP border routers and branch SD-WAN appliances natively support hardware-assisted BFD echo timers down to 15ms intervals (3x 15ms = 45ms detection), custom AS-path prepending, and transparent pass-through of customer BGP extended community strings (65001:100 through 65001:450) directly into Epic EHR VRFs.

SECTION 4: GOOGLE CLOUD INTERCONNECT & QUINCY EPIC EHR PEERING

Section 4.1 - Redundant 100Gbps Google Cloud Dedicated Interconnect (us-west1)

We provision 2x 100Gbps Google Cloud Dedicated Interconnect connections across POP-SEA-01 (Seattle Westin) and POP-PDX-01 (Portland Pittock) with physically diverse optical paths into Google Cloud us-west1, meeting GCP's 99.99% Interconnect SLA.

Section 4.2 - Cloud Interconnect MTU (8,896 Bytes) & Direct Multi-Cloud Routing

All 10Gbps optical wavelengths support 9,216-byte Jumbo Frames end-to-end, while Google Cloud Dedicated Interconnect VLAN attachments operate at GCP's maximum 8,896-byte hardware MTU with automatic MSS clamping.

Section 4.3 - Quincy Epic EHR Direct Optical Enclave (0.4ms Campus Latency)

POP-QCY-01 in Quincy, WA sits within 400 meters of the Primary Epic EHR hosting enclave, delivering 0.4ms intra-campus latency and enabling all 45 clinics across WA, OR, ID, and CA to achieve 1.5ms to 10.4ms RTD (well below the 12.0ms SLA mandate).

Section 4.4 & 4.5 - Synthetic Epic Hyperspace Probes & RPKI Route Security

Synthetic HTTPS/ICA probes execute every 250ms against Epic Hyperspace VIPs, triggering automatic BGP local-preference failover to Google Cloud us-west1 DR in <750ms. All BGP sessions enforce RPKI origin validation and TCP-AO authentication.

Section 5.1 - uCPE Virtual Network Function (VNF) Hosting Specifications

Branch uCPE-Pro appliances include dedicated KVM/Docker compute blades (up to 8 vCPUs, 32GB ECC RAM, 512GB NVMe) capable of hosting customer-managed clinical IoT packet brokers (4 vCPU / 16GB RAM) at zero additional MRC on a 36-month term.